

# SINAN YAVUZ

Seattle, WA

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## EDUCATION

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### University of Wisconsin - Madison

September 2021 - May 2026

Ph.D. in Quantitative Methods - Educational Psychology

Dissertation Title: *Subgroup Achievement Estimation for Large Scale Assessments: Small Area Bayesian Dynamic Borrowing Model*

### Hacettepe University, Ankara, Turkey

October 2012 - June 2021

Ph.D. in Educational Measurement and Evaluation

Dissertation Title: *Investigating Growth Models of Academic Achievement Scores by Maximum Likelihood And Bayesian Estimations*

### University of Wisconsin - Madison

January 2018 - September 2021

M.S. in Quantitative Methods - Educational Psychology

Thesis Title: *A Comparison of the Predictive Performance of Model Averaging and Model Selection Methods*

### Gazi University, Ankara, Turkey

September 2004 - August 2009

B.S. and M.S. in Biology Education

## WORK EXPERIENCE

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### Amazon, People eXperience and Technology (PXT)

Sep 2025 - Present

Senior Research Scientist

Seattle, WA

- Owned the nonresponse-bias correction for the worldwide Operations employee survey end to end (1.25M associates, running in production). Designed the entropy-balancing algorithm, authored the methodology paper, cleared internal science peer review, and cut the estimation pipeline from a week to under 15 minutes on AWS Glue Spark.
- Investigated the manager-level survey results still flagged as unrepresentative after weighting and decomposed the flag rate into coverage, targeting, and weighting causes. The engineering team fixed the question targeting and implemented my proposed weight-reliability guardrail in production.
- Led the feature-engineering and machine-learning work to improve nonresponse-bias correction: built a feature-discovery pipeline across the enterprise HR data estate for 1.1M employees, with automated inventory, leakage and redundancy filtering, and tree-based screening with stability selection, raising outcome-model AUC from 0.61 to 0.90.
- Designed the two-arm estimator for a company-wide employee-choice survey (~140K eligible) feeding a senior-leadership decision: a census arm and a probability-sample arm combined through selection weights and entropy balancing to a common frame.
- Re-architected the Bedrock LLM free-text classification pipeline (400K+ responses per run), cutting a 50+ hour run to about an hour at 99.9% completion and keeping a leadership deliverable on schedule. The pipeline has since grown into a multi-team chatbot effort. Built the 90-question accuracy benchmark for the internal LLM insights tool.
- Analyzed item-refusal and nonresponse rates across the operations and corporate populations for a reporting-threshold decision, correcting an earlier cross-organization comparison and recommending a minimum-respondent floor; delivered interactive charts for the working session.
- Stress-tested the weighting pipeline under population-exclusion scenarios on Glue clusters to clear a launch-blocking question ahead of a major rollout.

### American Institutes for Research

Sep 2021 - Sep 2025

Researcher - Full Time ([link](#))

Remote

- Led 10+ end-to-end research projects from study design to statistical modeling and dissemination, covering synthetic data simulation, small area estimation, multilevel multiple imputation, and imputation-based bias correction.

- Developed and implemented a novel Bayesian hierarchical model (SABDB) for subgroup estimation in small domains, maintaining 0.93 interval coverage under both homogeneity and heterogeneity where the standard HBSAE model collapsed to 0.50; in PISA applications, produced intervals 28% narrower that agreed with direct survey evidence for 97% of the smallest domains. Applied in a 2026 *Large-scale Assessments in Education* study of expanded state subgroup reporting.
- Led bias correction for the NAEP Full Population Estimates project, modeling counterfactual outcomes for the 2–5% of students excluded or nonresponding in every state and jurisdiction via multiple imputation and plausible values; NCES released the results as the official 2022 FPE report.
- Replicated unexpected 2022 NAEP results for the Los Angeles and Cleveland districts and ran Oaxaca–Blinder decompositions to identify what drove the change; findings delivered to NCES ahead of public release.
- Quantified the link between absenteeism and the 2022 NAEP declines in grades 4 and 8 mathematics and reading (absence-related gaps of roughly 5–20 scale-score points); presented at NCME 2024.
- Led generation of the first public-use synthetic NAEP dataset, replicating stratified sampling structures, survey weights, plausible values, item parameters, and missingness; released as an official NCES data product (NCES 2024-013).
- Ran all analyses for a two-year evaluation of ESSER-funded tutoring for Milwaukee Public Schools, building the participation and outcome metrics reported to the district across multiple student groups.
- Delivered statistical analyses (HLM, SEM, IRT, MCMC, machine learning) across three national and international assessment programs (NAEP, TIMSS, PIAAC), informing multiple NCES technical reports and memos.
- Co-developed a generative AI chatbot integrating R-based modeling and NLP to support large-scale assessment data interpretation; presented at NCME and AERA.
- Authored statistical functions in the EdSurvey R package (70,000+ CRAN downloads), the standard tool for NAEP and TIMSS analysis, and led six national training workshops.
- Delivered over 20 presentations and 6 trainings at national and international venues (NCME, AERA, IMPS), covering Bayesian modeling, small area estimation, synthetic data, and generative AI applications in education research.
- Chaired the Psychometricians' Community of Practice at AIR, promoting collaboration, mentoring, and professional development across 100+ members.

### **American Institutes for Research**

May 2019 - Aug 2021

*Intern Researcher - Part time*

- Led end-to-end research studies on school-level and contextual variable missingness in large-scale assessment data, including design, simulation, analysis, and presentation.
- Applied and compared multiple imputation techniques under different missing data mechanisms, including non-random missingness and multistage testing structures.
- Evaluated the impact of plausible value generation and missing data strategies on scale score validity and subgroup comparability.
- Co-authored and presented research findings at multiple national conferences, including NCME and AERA, covering imputation methodology, synthetic data generation, and measurement invariance.

### **Wisconsin Center for Education Research, UW–Madison**

August 2019 - June 2022

*Project Assistant*

*Madison, WI*

- Co-developed the Bayesian Dynamic Borrowing (BDB) methodology for multilevel and longitudinal educational data in an IES-funded project, published in *Psychometrika*.
- Designed and executed Monte Carlo simulations on high-throughput computing infrastructure to evaluate model performance across varying conditions.
- Contributed to ShinyBHB, the companion R Shiny app for interactive implementation of the BDB method.

### **Wisconsin Center for Education Research, ILDL**

September 2017 - August 2019

*Project Assistant*

*Madison, WI*

- Ran all analyses for an NSF-funded classroom experiment ([Interactive Learning and Design Lab](#)) across multiple schools, with classrooms assigned to a new curriculum or a control condition; fit multilevel models for the treatment comparisons, published in *IJCSSL* (2023) and *CSSL* (2019).
- Designed several of the student questionnaires and deployed them via Qualtrics; conducted descriptive analyses, t-tests, regression modeling, and hierarchical linear modeling.

## University of Wisconsin - Madison

August 2016 - September 2017

*Research Associate*

- Applied Bayesian growth curve models to analyze longitudinal academic achievement data.
- Developed a novel multiple imputation Bayesian model averaging (miBMA) approach to address model uncertainty in missing data imputation.

## Hacettepe University, Ankara Turkey

February 2013 - August 2016

*Research Assistant*

- Served as Student Government President for the Educational Sciences Institute and Department, representing over 3,000 graduate students in policy discussions and academic initiatives.
- Conducted a linking and equating project for the Measurement, Selection, and Placement Center (ÖSYM) in Turkey for a dataset with over 1.7M participants.
- Actively collaborated in multiple research studies on item analysis, reliability, differential item functioning, and validity, and provided statistical consulting support to graduate students and faculty members.

## Murat Educational Institution, Ankara, Turkey

September 2013 - June 2016

*Instructor (Part-time)*

- Designed curriculum and taught 1,000+ hours of undergraduate and graduate educational measurement and assessment coursework over three years, plus Learning and Developmental Psychology courses.

## TECHNICAL STRENGTHS

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### Software

- Python, SQL, R (package author; github: [Sinan-Yavuz](#)), Stan, JAGS, Spark / AWS Glue, Amazon Bedrock LLM APIs, R Shiny, Mplus, SAS, SPSS, L<sup>A</sup>T<sub>E</sub>X

### Statistical Methods

- Bayesian Hierarchical Modeling, Bayesian Model Averaging, Small Area Estimation, Missing Data and Multiple Imputation, Complex Survey Sampling and Weighting, Nonresponse Bias Correction, Causal Inference and Reweighting (Entropy Balancing, IPW), Experimental Design and Multilevel Analysis of Field Experiments, Machine Learning (Tree Ensembles, Feature Selection), LLM Pipelines and Evaluation, Multilevel Modeling, Structural Equation Modeling, Item Response Theory, Classical Test Theory, Differential Item Functioning, Test Equating, High Throughput Computing.

## GRANT SUBMISSIONS

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Title: *A Century of Significance: Considering the National Center for Education Statistics' Statistical Standards*

IES Statistical and Research Methodology in Education

Status: Pending

Role: Co-PI (PI: Rachel Carroll, PhD, American Institutes for Research)

Title: Bayesian Methods Training for Educational Research

National Center for Educational Statistics

Status: Pending

Role: Co-PI (PI: Christopher Thompson, PhD, Texas A&M University)

Title: Procurement of Research Evaluation and Statistics Task Orders

Institute of Education Sciences

Status: Ongoing

Role: Key Personnel

## PEER REVIEWED PUBLICATIONS

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1. Bamat, D., **Yavuz, S.**, & Grekov, P. (2026). An appraisal of small area estimation methods to explore expansion of state subgroup reporting on NAEP. *Large-scale Assessments in Education*, 14(1), 32. <https://doi.org/10.1186/s40536-026-00306-w>
2. **Yavuz, S.**, Bailey, P., Zhang, T., and Sikali, E. (2024). Simulated 2015 National Assessment of Educational Progress (NAEP) Grade 8 Mathematics Public-Use Methodological Data Files Data Companion (NCES 2024-013). *National Center for Education Statistics, Institute of Education Sciences, U.S. Department of Education. Washington, DC.*
3. Puntambekar, S., Gnesdilow, D., & **Yavuz, S.** (2023). Understanding the Effect of Differences in Prior Knowledge on Middle School Students' Collaborative Interactions and Learning. *Intern. J. Comput.-Support. Collab. Learn.* <https://doi.org/10.1007/s11412-023-09405-0>
4. Kaplan, D., Chen, J., Lyu, W. & **Yavuz, S.** (2023). Bayesian Historical Borrowing with Longitudinal Large-Scale Assessments. *Large-scale Assess Educ* 11, 2. <https://doi.org/10.1186/s40536-022-00140-w>
5. Kaplan, D., Chen, J., **Yavuz, S.** & Lyu, W. (2022). Bayesian Dynamic Borrowing of Historical Information with Applications to the Analysis of Large-Scale Assessments. *Psychometrika*, 1-30.
6. Kaplan, D. & **Yavuz, S.** (2019). An approach to addressing multiple imputation model uncertainty using Bayesian model averaging. *Multivariate behavioral research*, 55(4), 553–567. <https://doi.org/10.1080/00273171.2019.1657790>
7. Dogan, N., Hambleton, R. K., Yurtcu, M. & **Yavuz, S.** (2018). The Comparison of Differential Item Functioning Predicted Through Experts and Statistical Techniques. *Cypriot Journal of Educational Science*. 13(2), 375-384.
8. Ozdemir, A., Yakar, L., & **Yavuz, S.** (2017). Examination of Mathematics Final Grade Components: Repealing The Performance Tasks. *Usak University Journal of Educational Research*, 3(1), 74-91.
9. **Yavuz, S.**, Odabas, M. & Ozdemir, A., (2016). Effect of Socio-Economic Status on Students' TEOG Mathematics Achievement. *Journal of Measurement and Evaluation in Education and Psychology*, 2016;7(1).
10. Sari, S., Kilic, N., Kurt, G., Dogulu, F., Ceviker, N., Oz, Y., Donmez, U., Yildirim, Z. & **Yavuz, S.** (2016) Investigation of Serum Levels of Inwardly Rectifying and Two-Pore Potassium Channel Proteins in Benign and Malign Brain Tumors. *Turkiye Klinikleri Journal of Medical Sciences*. 2016;36(1):7-13.

## PREPRINTS

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1. **Yavuz, S.**, & Kaplan, D. (2026). Small area Bayesian dynamic borrowing: Adaptive subgroup estimation for large-scale educational assessments. *arXiv*. <https://doi.org/10.48550/arXiv.2608.07708>

## BOOK CHAPTERS

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1. Liao, Y., Bailey, P., & **Yavuz, S.** Models. In Bailey, P. and Zhang, T. (eds.), *Analyzing NCES Data Using EdSurvey: A User's Guide*. [https://naep-research.airprojects.org/Portals/0/EdSurvey\\_A\\_Users\\_Guide/\\_book/models.html](https://naep-research.airprojects.org/Portals/0/EdSurvey_A_Users_Guide/_book/models.html)
2. Bailey, P., Cohen, M., & **Yavuz, S.** Statistical Methods Used in EdSurvey. In Bailey, P. and Zhang, T. (eds.), *Analyzing NCES Data Using EdSurvey: A User's Guide*. [https://naep-research.airprojects.org/Portals/0/EdSurvey\\_A\\_Users\\_Guide/\\_book/methods.html](https://naep-research.airprojects.org/Portals/0/EdSurvey_A_Users_Guide/_book/methods.html)
3. **Yavuz, S.**, Forkner M., Gnesdilow D., Dornfeld-Tissenbaum C. L., Kim J. & Puntambekar S. (2019). Understanding the Effect of Group Variance on Learning. In Lund, K., Niccolai, G. P., Lavoué, E., Hmelo-Silver, C., Gweon, G., & Baker, M. (Eds.), *A Wide Lens: Combining Embodied, Enactive, Extended, and Embedded Learning in Collaborative Settings*, 13th International Conference on Computer Supported Collaborative Learning (CSCL) 2019, Volume 1 (pp. 520-525). Lyon, France: International Society of the Learning Sciences.

## TECHNICAL REPORTS

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1. Adams, E., **Yavuz, S.**, Chamberlain, R. (2025) Milwaukee Public Schools ESSER Evaluation: Tutoring Services Descriptive Analysis of ESSER-Funded Tutoring Services in 2022-23 School Year [Technical Report]. American Institutes for Research. Prepared for the Milwaukee Public Schools.
2. Adams, E., **Yavuz, S.**, Chamberlain, R. (2024) Milwaukee Public Schools ESSER Evaluation: Tutoring Services Final Report on ESSER-Funded Tutoring Services (2022-23 and 2023-24) [Technical Report]. American Institutes for Research. Prepared for the Milwaukee Public Schools.
3. **Yavuz, S.** (2024). Summary of the FPE Comparisons in 2022 NAEP Between the Operational Conditioning Model Based FPE and the Revised Conditioning Model: Mathematics and Reading at Grades 4 and 8 [Technical memo]. American Institutes for Research. Prepared for the National Center for Education Statistics.
4. **Yavuz, S.** (2023). Estimation of Standard Deviation and Error of Standard Deviation for Full Population Estimates Results [Technical memo]. American Institutes for Research. Prepared for the National Center for Education Statistics.
5. National Center for Education Statistics. (2023). Full population estimates 2022 NAEP grade 4 and 8, mathematics and reading results report. U.S. Department of Education.  
<https://nces.ed.gov/nationsreportcard/about/fpe.aspx>
6. **Yavuz, S.**, Zhang, T., Bailey, P., & Kim, Y. Y. (2022). Replication and Investigation of Los Angeles TUDA Grade 8 NAEP Reading Results [Technical memo]. American Institutes for Research. Prepared for the National Center for Education Statistics.
7. **Yavuz, S.**, Bai, Y., Zheng, X., & Broer, M. (2022). *Multiple imputation approach to missingness at the school level: A comparison study* (Research Study No. 43). AIR-ESSIN Task 14.
8. **Yavuz, S.**, Zhang, T., Bailey, P., & Kim, Y. Y. (2022). Replication and investigation of Cleveland TUDA Grade 4–8 NAEP reading and math results [Technical memo]. American Institutes for Research. Prepared for the National Center for Education Statistics.

## UNPUBLISHED PAPERS

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1. **Yavuz, S.** (2021). A Comparison of the Predictive Performance of Model Averaging and Model Selection Methods. Unpublished Master's Thesis. University of Wisconsin - Madison, Educational Psychology, Madison, Wisconsin.
2. **Yavuz, S.** (2021). Investigating Growth Models of Academic Achievement Scores by Maximum Likelihood and Bayesian Estimations. Unpublished Doctoral Dissertation. Hacettepe University, Graduate School of Educational Sciences, Ankara.

## SOFTWARE

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1. Bailey, P., Emad, A., Huo, H., Lee, M., Liao, Y., Lishinski, A., Nguyen, T., Xie, Q., Yu, J., Zhang, T., Buehler, E., Lee, S., Webb, B., Fink, T., & **Yavuz, S.** (2024). *EdSurvey: Analysis of NCES Education Survey and Assessment Data*. R package version 4.0.8, <https://www.air.org/project/nces-data-r-project-edsurvey>. 70,000+ CRAN downloads.
2. **Yavuz, S.** *SABDB: Small Area Bayesian Dynamic Borrowing*. R implementation. <https://github.com/Sinan-Yavuz/SABDB>
3. **Yavuz, S.** *miBMA: Multiple imputation under Bayesian model averaging*. R implementation. <https://github.com/Sinan-Yavuz/miBMA>
4. Balancing-weights package (internal Python, Amazon). Entropy balancing and inverse-propensity weighting with full balance diagnostics and print-ready plots.
5. Contributor, *ShinyBHB*: R Shiny application for Bayesian historical borrowing in large-scale assessments.

## PRESENTATIONS

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1. **Yavuz, S.** (2026). An Area-Level Small Area Bayesian Dynamic Borrowing Model. Accepted at the National Council on Measurement in Education, 8-11 April 2026.
2. **Yavuz, S.** (2025). An Innovative Way of Estimating Subgroups Mean Estimates: SABDB. Presented at the National Council on Measurement in Education, 23-26 April 2025, Denver, CO.
3. Grekov, P., Bamat, D., & **Yavuz, S.** (2025). Expanding NAEP Subgroup Reporting with Random Forests. Presented at the National Council on Measurement in Education, 23-26 April 2025, Denver, CO.
4. **Yavuz, S.**, Webb, B., Zhang, T., Liao, Y., & Sikali, E. (2024). Generative AI chatbot for large-scale assessment data analysis. Presented at the National Council on Measurement in Education, 11-14 April 2024, Philadelphia, PA.
5. **Yavuz, S.**, & Kim, Y. Y. (2024). Potential Effect of Absenteeism on Declines of Students' Performance in NAEP 2022. Presented at the National Council on Measurement in Education, 11-14 April 2024, Philadelphia, PA.
6. Okay, S., **Yavuz, S.**, & Gelbal, S. (2024). Applying LASSO regression to understand house effect on PIRLS 2021 Turkey results. Presented at the National Council on Measurement in Education, 11-14 April 2024, Philadelphia, PA.
7. Kim, Y. Y., **Yavuz, S.**, & Zheng, X. (2024). Deeper dive into the relationship between absent rates and NAEP performance. Presented at the National Council on Measurement in Education, 11-14 April 2024, Philadelphia, PA.
8. Siegel, D., Zhang, T., Bailey, P., & **Yavuz, S.** (2024). Estimating Student Achievement in Large-Scale Assessments: Can Latent Regression Substitute Plausible Values? Presented at the National Council on Measurement in Education, 11-14 April 2024, Philadelphia, PA.
9. Lyu, W., Ye, S., Chen, J., **Yavuz, S.**, & Kaplan, D. (2024). ShinyBHB: A Shiny App for Bayesian Historical Borrowing in Large-Scale Assessments. Presented at the National Council on Measurement in Education, 11-14 April 2024, Philadelphia, PA.
10. Zhang, T., Liao, Y., Bailey, P., Webb, B., **Yavuz, S.**, & Sikali, E. (2024). Generative AI chatbot as NAEP helpdesk. Presented at the National Council on Measurement in Education, 11-14 April 2024, Philadelphia, PA.
11. **Yavuz, S.**, Zhang, T., Bailey, P. & Sikali, E. (2023). A Use Case of Simulating NAEP-like Data. Presented at the National Council on Measurement in Education, 12-15 April 2023, Chicago, IL.
12. Kaplan, D., Chen, J., Lyu, W., & **Yavuz, S.** (2022). Bayesian Dynamic Borrowing With Longitudinal Large-scale Assessment Data. International Meeting of the Psychometric Society, July 11th - July 15th, 2022, Bologna, Italy.
13. Kaplan, D., Chen, J., **Yavuz, S.**, & Lyu, W. (2022). Bayesian Dynamic Borrowing with Applications to Large-Scale Educational Assessments. International Society for Bayesian Analysis, June 26th - July 1st, 2022, Montreal, Canada.
14. **Yavuz, S.**, Zhang, T., & Bailey, P. (2022). Generating Synthetic Large-Scale Assessment Data. Presented at the National Council on Measurement in Education, 22-25 April 2022, San Diego, CA.
15. Zheng, X., **Yavuz, S.**, Bai, Y., & Broer, M. (2022). Comparing Imputation Methods for School-Level Variables in Large-Scale Assessments. Presented at the National Council on Measurement in Education, 22-25 April 2022, San Diego, CA.
16. Gocer Sahin., S., Bishop., K., L., & **Yavuz., S.** (2022). Measurement Invariance of English Proficiency Test on U.S. and International ELLs. Presented at the National Council on Measurement in Education (NCME), 22-25 April 2022, San Diego, CA.
17. Zhang, T., Bailey, P., **Yavuz, S.**, & Huo, H. (2021). Imputations for Large-Scale Assessment Contextual Data: Is Recreation of Plausible Values Necessary? Presented at the National Council on Measurement in Education, 8-11 June 2021, Virtual.

18. **Yavuz, S.**, Zheng, X., Bai, Y., & Broer, M. (2021). Multiple Imputation Approach to Missingness at the School Level: A Comparison Study. Presented at the National Council on Measurement in Education, 8-11 June 2021, Virtual.
19. **Yavuz, S.**, Zheng, X., & Kim, Y. Y. (2021). Using Multiple Imputations to Handle Non-Random Missing in Multidimensional Multistage Testing. Presented at the National Council on Measurement in Education, 8-11 June 2021, Virtual.
20. **Yavuz S.**, Kaplan, D., Chen, J., & Lyu, W. (2021). On the development and extension of Bayesian historical borrowing to single-level and multilevel models. Invited Presentation at Open Science Grid All-Hands Meeting, 1-5 March, 2021, Virtual.
21. **Yavuz S.** & Kaplan, D. (2020). Comparison of Model Averaging and Model Selection Methods. Presented at International Meeting of Psychometric Society, 14-17 July 2020, Virtual.
22. **Yavuz S.** & Kaplan, D. (2019). A Comparison of the Predictive Performance of Model Averaging and Model Selection Methods. Presented at The Ideas In Testing Research Seminar, November 8th, 2019, Chicago, IL.
23. Kaplan, D. & **Yavuz, S.** (2019.) An Approach to Addressing Multiple Imputation Model Uncertainty Using Bayesian Model Averaging. Presented at the International Meeting of the Psychometric Society. 15-19 July, 2019, Santiago, Chile.
24. Kaplan, D. & **Yavuz, S.** (2018). An Approach to Addressing Multiple Imputation Model Uncertainty Using Bayesian Model Averaging. Presented at the 8th Meeting of the European Association for Methodology. July 27th 2018, Jena, Germany.
25. **Yavuz S.** & Kaplan, D. (2018). Comparison of Frequentist and Bayesian Model Averages. Presented at the International Meeting of Psychometric Society, 10-13 July 2018, New York, NY.
26. Kaplan, D. & **Yavuz, S.** (2017). A Really, Truly, “Bayesianly” Proper Approach to Multiple Imputation. Paper presented at the annual meeting of the Society of Multivariate Experimental Psychology, 4-7 October 2017, Minneapolis, MN.
27. Ozdemir A., **Yavuz S.** (2017). Comparison Of Bayesian Model Averaging And Linear Regression Prediction Powers. Presented at the 9th International Congress of Educational Research, 4-11 May 2017, Ordu, Turkey.
28. Kuru, N., Kosan, Y. & **Yavuz S.** (2016). Investigating the relationship between Parents Attitude and Kids Academic Self Respect. Presented at the International Conference on Educational Sciences, 21-24 April 2016, Antalya, Turkey.
29. Dogan N., Hambleton R.K., Yurtcu M. & **Yavuz S.** (2016). The Comparison of Differential Item Functioning Predicted Through Experts and Statistical Techniques. Presented at the 8th World Conference on Educational Sciences, WCES, 4-6 February 2016, Madrid, Spain.
30. Kumlu G., **Yavuz S.** & Dogan N. (2015). Comparing Pairwise Comparison and Rank-Order Judgments Scaling On Teacher Competences. Presented at the International Meeting of Psychometric Society, IMPS, 12- 16 July 2015, Beijing, China.
31. Dogan N., Hambleton R.K., Yurtcu M. & **Yavuz S.** (2015). Experts’ Prediction Powers of Differential Item Functioning Levels of Math and Turkish Tests Items. Presented at the International Congress on Education for the Future: Issues and Challenges, ICEFIC, May 13-15, 2015, Ankara, Turkey.
32. Yakar L, Odabas M. & **Yavuz S.** Investigating Distance Education Teaching Certificate Program Scores with Respect to Some Variables. Presented at the International Distance Education Conference, ID-EC 18-20 December 2014, Doha, Qatar.
33. Yildiztekin B., Kumlu G., **Yavuz S.** & Yakar L. (2014). Investigation of 8th Grade Students Math Written Exam Scores comparing to TEOG Exam Scores. Presented at the International Conference on New Horizons in Education, 25-27 June 2014, Paris, France.
34. Odabas M. & **Yavuz S.** (2014). Examining Teog Mathematics Achievement and Social Economic Status Relationship by HLM. Presented at the Congress of the Measurement and Evaluation in

Education and Psychology, EPOD, 9-13 June 2014, Ankara, Turkey.

35. Yakar L, Ozdemir A. & **Yavuz S.** (2014). Impact of 8th Grade Math Ratings Scores Components on Students Final Scores. Presented at the Congress of the Measurement and Evaluation in Education and Psychology, EPOD 9-13 June 2014, Ankara, Turkey).
36. Yakar L. & **Yavuz S.** (2014). Differential Item Functioning According to Test Booklets. Presented at the Congress of the Measurement and Evaluation in Education and Psychology, EPOD 9-13 June 2014, Ankara, Turkey.

## WORKSHOPS

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1. **Yavuz, S.**, Sikali, E., Zhang, T., & Bailey, P. (2024, April). Using R and Generative AI for NAEP Data Analysis. The American Educational Research Association (AERA) Workshop, Philadelphia, PA.
2. Sikali, E., Zhang, T., Bailey, P.D., **Yavuz, S.**, and Webb, B. (March 28, 2023) Analyzing NAEP and TIMSS Data in R Using Latent Regression. NCME 2023 Virtual Research Learning Series 2023.
3. Sikali, E., Bailey, P.D., Zhang, T., Lee, M., **Yavuz, S.**, and Hooper, M. (June 2nd, 2022) Analyzing NAEP and TIMSS Data Using R. AERA 2022 Virtual Research Learning Series 2022.
4. Zhang, T. Sikali, E., Bailey, P.D., Lee, M., and **Yavuz, S.** (2022 April) Analyzing NAEP/TIMSS Data with Direct Estimation in R, Theory and Practice. Presented at the National Council on Measurement in Education (NCME), San Diego, CA.
5. Bailey, P.D., Zhang, T., Mamedova S., Pawlowski, E., Sikali, E., Lee, M., Buehler, E., and **Yavuz, S.** (March 22nd, 2022) Analyzing PIAAC data using the R EdSurvey package. Pre-Conference Workshops at the International PIAAC Research Conference 2022.
6. Zhang, T. Sikali, E., Bailey, P.D., Lee, M., Buehler, E., and **Yavuz, S.** (2021 October) Analyzing NAEP and TIMSS Data Using R: 2021 EdSurvey Large-Scale Data Analysis Training Series. NCES workshop series presented virtually.

## TRAININGS/PROFESSIONAL DEVELOPMENT

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1. Morgridge Entrepreneurial Bootcamp (MEB) 2022, University of Wisconsin-Madison. June 6 - 10, 2022.

## PROFESSIONAL MEMBERSHIP AND VOLUNTEER POSITIONS

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1. AERA NAEP Studies SIG Newsletter Editor (2023 - Present)
2. Member of the Psychometric Society, AERA, and NCME

## AWARDS

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1. TUBITAK (Scientific and Technological Research Council of Turkey) Dissertation Grant (2016-2017), \$21,600
2. UW-Madison School of Education Travel Award
3. Jenny Alexander Perseverance Fellowship (2024), \$3,500